



KENYATTA UNIVERSITY
EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE AND
BACHELOR OF EDUCATION

SMA 102: BASIC MATHEMATICS

1ST SEMESTER 2021/2022

TIME: 2 Hours

INSTRUCTIONS: Attempt Question ONE and any other TWO

Question One (30 Marks)

a) Draw a Venn diagram representing $(A \cup B) - (A \cap B)$

(2 marks)

b) Give the truth table of the proposition give by $\sim (p \wedge q) \Leftrightarrow (\sim p) \vee (\sim q)$.

(3 marks)

c) State the converse, inverse and the contrapositive of

'if a triangle is equiangular, then it is equilateral'

(3 marks)

d) If α and β are the roots of the quadratic equation $2x^2 - 4x - 3 = 0$, find the quadratic equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

$$\alpha + \beta = 2, \quad \alpha - \beta = -3, \quad \alpha\beta = -\frac{3}{2}$$

$$\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{2^2 - 2(-\frac{3}{2})}{-\frac{3}{2}} = \frac{4 + 3}{-\frac{3}{2}} = -\frac{14}{3}$$

$$\frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1$$

e) Determine the line of symmetry and minimum or maximum point of the graph $f(x) = 2 - 2x - x^2$ without drawing the curve.

(4 marks)

f) When the polynomial $f(x) = x^3 + 2x^2 - 5ax - 7$ is divided by $x + 1$, the remainder is R_1 . When the polynomial $h(x) = x^3 + ax^2 - 12x + 6$ is divided by $x - 2$, the remainder is R_2 . If $2R_1 + R_2 = 6$, find the value of a .

$$R_2 = 6 - 2R_1$$

$$\text{at } x = -1 \text{ be } P \text{ and } a = 2$$

$$P = 6 - 2 \cdot 2 = 2$$

$$\frac{3}{2} \sqrt{7} \quad \frac{7}{2} \pm 3 \text{ rem } 4$$

$$7 = 2(3) + 1$$

g) Solve $z^2 + 1 = 0$

(4 marks)

h) Find $\text{Re}(z)$ and $\text{Im}(z)$ given that $z = \frac{2-i\sqrt{3}}{i+\sqrt{3}}$

(3 marks)

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i) Prove that $\sqrt{6}$ is an irrational number.

(4 marks)

Question Two (20 Marks)

a) The function $f(x) = x^3 - 4x^2 + ax + b$ is such that it is divisible by $x - 3$ and when divided by $x - 1$, there is a remainder of 4. Find the values of a and b .

$$\begin{array}{l} 8 \div 2 = 4 \text{ rem } 0 \\ 8 \div 3 = 2 \text{ rem } 2 \end{array}$$

$$\begin{array}{l} 2 \times 4 = 8 \\ (3 \times 2) + 2 = 8 \end{array}$$

(4 marks)

b) The roots of the quadratic equation $2x^2 + 7x - 3 = 0$ are α and β . Find the quadratic equation whose roots are $\alpha + \frac{5}{\beta}$ and $\beta + \frac{5}{\alpha}$.

$$\begin{array}{l} \alpha + \beta + 3 + \frac{5}{\beta} \\ \beta + 5 + 5 + \frac{25}{\alpha\beta} \end{array}$$

(5 marks)

c) In a class of 100 students, 35 like geography, 45 like History and 10 like both. How many like either of them and how many like neither?

$$G \cup H$$

$$G \cup H (G \cup H)^c$$

(3 marks)

d) Find the set of all values of x such that $12x^2 < x + 6$.

$$\begin{array}{l} 12x^2 < x + 6 \\ 12x^2 - x - 6 < 0 \end{array}$$

(4 marks)

e) Simplify $(A^c \cap B) \cup (A^c \cap B^c) \cup (A \cap B)$

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Question Three (20 Marks)

a) Solve the equation $x^3 - 2x^2 - 4x + 8 = 0$.

(4 marks)

b) I have nine paintings and have room to display only four of them at a time on my wall. How many different ways could I do this?

(2 marks)

c) Find the coefficient of x^5 in the expansion of $(4y - 3x)^{15}$

(3 marks)

d) There are 16 boys and 12 girls in a class. How many ways can a committee of 4 people be formed if there must be at least 2 girls on the committee?

(4 marks)

e) Use the binomial theorem to expand $(8 - 3x)^{\frac{1}{3}}$ in ascending powers of x up to the term in x^3 . Use your expansion, with suitable value of x , to obtain an approximation to $\sqrt[3]{7.7}$ correct to 3 decimal places.

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(5 marks)

- f) Find the number of arrangements of the letters of the word PARRAMATTA.

(2 marks)

Question Four (20 Marks)

- a) Use Boolean algebra to simplify the following.

i) $(p \wedge q) \vee (\sim p \wedge q) \vee (p \wedge \sim q)$

(3 marks)

ii) $\sim(p \Rightarrow q) \vee p$

(4 marks)

- b) Show whether $(p \Rightarrow q) \wedge \sim q \Rightarrow \sim p$ is a tautology, a contradiction or neither.

(3 marks)

- c) (a) Find the contrapositive of the converse of $p \Rightarrow \sim q$.

(2 marks)

- (b) Write the negation of $\sim p \wedge \sim q$ in its simplest form.

(2 marks)

- d) Translate the following argument into symbolic form and using truth table, determine the validity of the argument;

If rains do not come, or no one works then the crops will fail.
The crops have failed.
Therefore, the rains did not come.

(6 marks)

Question Five (20 Marks)

- a) By giving a counter example, disapprove, the statement

$$\forall x \in \mathbb{R}, (x^2 > 0).$$

(2 marks)

- b) Prove by induction that

$$2 + 4 + 6 + \dots + 2n = n(n + 1).$$

(4 marks)

- c) (a) Express the complex number $\frac{1+2i}{-4+3i}$ in the form $a + bi$, where $a, b \in \mathbb{R}$.

(2 marks)

(b) Express the complex number $z = -5 + 5i$ in polar form

(3 marks)

d) (a) By using De Moivre's theorem, express $(1 + i)^{10}$ in the form $a + bi$, where $a, b \in \mathbb{R}$.

(4 marks)

(b) Find the square roots of i .

(5 marks)

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